

# SU(N) Yang-Mills Theory and Scalar Fields

*Perimeter Scholars International, Quantum Field Theory II, Tutorial 8; 11 worked parts.*

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

## 1. Coupling to Scalar Fields

Let  $\phi_i(x)$  be a scalar field in a representation  $r$  of  $SU(N)$ , with  $i = 1, \dots, \dim r$ .

- a) Determine the dimension of the group  $SU(N)$ .

**Solution.** An  $N \times N$  complex matrix has  $2N^2$  real dimensions. The hermitian condition,  $U_{ik}^* U_{jk} = \delta_{ij}$ , is  $N^2$  constraints. In addition,  $\det U = 1$  gives an extra constraint. Therefore,  $\dim SU(N) = N^2 - 1$ .

- b) Determine the dimension of the fundamental representation of  $SU(N)$ .

**Solution.** The fundamental representation of  $SU(N)$  is  $N \times N$  complex matrices, which has dimension  $N$ .

- c) The adjoint action is defined by

$$\text{Ad}_g(T^a) \equiv g T^a g^{-1}. \quad (1.1)$$

Thus there is a  $\dim G \times \dim G$  matrix  $C(g)$  such that

$$g T^a g^{-1} = C^{ab}(g) T^b. \quad (1.2)$$

Expand  $g$  and  $C(g)$  infinitesimally and use

$$[T_i, T_j] = i f_{ijk} T_k \quad (1.3)$$

to derive the adjoint generators

$$(T_i)_{jk} = i f_{ijk}. \quad (1.4)$$

**Solution.**

$$\text{Ad}(g)Y = (I + \epsilon X)Y(I - \epsilon X) = I + \epsilon[X, Y] = (I + \epsilon \text{ad}(X))Y \implies \text{ad}(X)Y = [X, Y]. \quad (1.5)$$

Taking  $X = T^a, Y = T^b$ , we have

$$\text{ad}(T^a)^{bc} T^c = i f^{abc} T^c \implies \text{ad}(T^a)^{bc} = i f^{abc}. \quad (1.6)$$

- d) Determine the dimension of the adjoint representation of  $SU(N)$ .

**Solution.** As we have seen above, the adjoint representation of the Lie algebra is a  $\dim G \times \dim G$  matrix. Therefore,

$$\dim \text{adj}(su(N)) = \dim SU(N). \quad (1.7)$$

Now consider the local transformation

$$\phi_i(x) \mapsto g_{ij}(x) \phi_j(x), \quad g(x) = \exp[i\alpha_m(x) T_r^m], \quad (1.8)$$

where  $T_r^m$  are the  $su(N)$  generators in the representation  $r$  and the  $\alpha_m(x)$  are spacetime-dependent group parameters. Require the covariant derivative to transform as

$$D_\mu \phi_i(x) \mapsto g_{ij}(x) D_\mu \phi_j(x), \quad (1.9)$$

and define, with matrix indices suppressed,

$$D_\mu \phi = \partial_\mu \phi - ie A_\mu \phi, \quad A_\mu(x) = A_\mu^m(x) T_r^m. \quad (1.10)$$

- e) Derive the finite gauge-transformation law of  $A_\mu(x)$  required by the covariance of  $D_\mu \phi$ .

**Solution.**

$$D_\mu \phi \rightarrow D'_\mu g \phi = (\partial_\mu g) \phi + g \partial_\mu \phi - ie A'_\mu g \phi = g(\partial_\mu \phi - ie A_\mu \phi), \quad (1.11)$$

which implies

$$A'_\mu = g A_\mu g^{-1} - \frac{i}{e} (\partial_\mu g) g^{-1}. \quad (1.12)$$

f) For infinitesimal  $\alpha_m(x)$ , show that

$$\delta A_\mu^a = \frac{1}{e} D_\mu \alpha_a, \quad D_\mu \alpha = \partial_\mu \alpha + ie[\alpha, A_\mu]. \quad (1.13)$$

**Solution.**

$$\begin{aligned} A_\mu^a R(T^a) &\rightarrow (I + i\alpha_b R(T^b)) A_\mu^a R(T^a) (I - i\alpha_b R(T^b)) + \frac{1}{e} (\partial_\mu \alpha_b) R(T^b) (I + i\alpha_a R(T^a)) \\ &= A_\mu^a R(T^a) + i\alpha_b [R(T^b), R(T^a)] A_\mu^a + \frac{1}{e} (\partial_\mu \alpha_b) R(T^b) \\ &= A_\mu^a R(T^a) + \frac{1}{e} D_\mu \alpha, \end{aligned} \quad (1.14)$$

from which we can recognise  $\alpha$  as the adjoint representation of  $su(N)$ .

g) Prove that the Lagrangian

$$\mathcal{L} = -(D_\mu \phi)^\dagger D^\mu \phi - m^2 \phi^\dagger \phi - \frac{\lambda}{2} (\phi^\dagger \phi)^2 \quad (1.15)$$

is invariant under the local scalar transformation and the gauge-field transformation found above.

**Solution.** It can be easily seen through the transformation of  $\phi$  and  $D_\mu \phi$ .

h) Extract every Feynman rule from this Lagrangian that contains at least one scalar line.

**Solution.** Expanding the covariant derivative, we have

$$|D_\mu \phi|^2 = |\partial_\mu \phi|^2 + ie A_\mu (\phi^\dagger \partial_\mu \phi - \phi \partial_\mu \phi^\dagger) + e^2 A_\mu^2 |\phi|^2, \quad (1.16)$$

and the Feynman rules can be read off.

i) Apply Noether's theorem to obtain the conserved currents.

**Solution.**

$$j_\mu^a = \frac{\partial \mathcal{L}}{\partial (\partial^\mu \phi)} \delta \phi^a + \frac{\partial \mathcal{L}}{\partial (\partial^\mu \phi^\dagger)} \delta \phi^{\dagger a} = i(D_\mu \phi)^\dagger R(T^a) \phi - i\phi^\dagger R(T^a) (D_\mu \phi). \quad (1.17)$$

## 2. Zero Modes of the Yang-Mills Lagrangian

Take the unfixed Yang-Mills Lagrangian

$$\mathcal{L} = -\frac{1}{2g^2} \text{Tr}(F^{\mu\nu} F_{\mu\nu}), \quad (2.1)$$

with the trace over adjoint indices and arbitrary  $N$  in  $SU(N)$ .a) Expand the  $F^2$  term, isolate the quadratic terms containing two derivatives, and write them as

$$\int d^4x A_\mu^a(x) D_x^{\mu\nu} A_\nu^a(x). \quad (2.2)$$

**Solution.**

$$\begin{aligned} S &= -\frac{1}{4g} \int d^d x F_{\mu\nu}^a F^{a\mu\nu} \\ &= -\frac{1}{2g} \int d^d x A_\nu (-\partial^2 \eta^{\mu\nu} + \partial^\mu \partial^\nu) A_\mu + \dots, \end{aligned} \tag{2.3}$$

where the interacting terms are suppressed.

b) Show that there is a nonzero field  $B_\mu$  satisfying

$$D_x^{\mu\nu} B_\nu(x) = 0. \tag{2.4}$$

*Hint: regard  $D^{\mu\nu}$  as an integral kernel and inspect its momentum-space representation.* Explain why this zero mode obstructs inversion of the quadratic operator before gauge fixing.

**Solution.** It is easy to see that any field of the form  $\partial_\mu \omega$  is the kernel of the operator  $D_a^{\mu\nu}$ . Therefore,  $D_a^{\mu\nu}$  is clearly not invertible and hence we need the Faddeev-Popov procedure.